

● Reproducible benchmark · A100-80GB

Axis vs PyTorch — 7B LoRA fine-tuning

Same model shape, same adapters, same GPU. Full-model 7B training doesn't fit on a single 80 GB GPU for either framework, so the single-GPU 7B benchmark is LoRA fine-tuning.

Model **Llama-2-7B shape** · dim4096 · 32L · 32h · mlp11008 · V32000

LoRA **r=16** · q,k,v,o,gate,up,down · ~40M trainable

seq **2048** · batch **1** · **bf16**

GPU **A100-80GB** · fresh container · 5 timed steps

Throughput

Latency

GPU memory

Higher is better — bar length scaled to the larger value.

■ Axis

3,833 tok/s

■ PyTorch

4,398 tok/s

best

FRAMEWORK	MS / STEP	TOKENS / S	GPU MEMORY	TRAINABLE
PyTorch (HF + PEFT + SDPA)	466	4,398	28.2 GB	40M
Axis	534	3,833	23.3 GB	40M

-17%

GPU memory vs PyTorch
(4.9 GB saved)

0.87×

Throughput vs PyTorch (466
vs 534 ms/step)

40M

Identical adapters trained by
both

READING THE RESULTS HONESTLY

- **Axis's win is memory.** The frozen 7B base carries no optimizer state and stays in bf16, so Axis fine-tunes in **17% less GPU memory** (23.3 GB vs 28.2 GB) — headroom for longer context or larger batches on the same card.
- **PyTorch is faster here (~15%).** A decade of hand-tuned CUDA kernels and PEFT integration is ahead on raw LoRA throughput at 7B. We publish that, not just the wins.
- **Apples-to-apples.** Both train the same ~40M adapters on the same 7 projections ($r=16$), same Llama-2-7B shape, same bf16 + flash attention, each in its own fresh container.
- **Full-model 7B training** needs ~112 GB for bf16 weights + grads + fp32 AdamW state — it doesn't fit on one 80 GB GPU for either framework; that's a multi-GPU (ZeRO / FSDP) benchmark, run separately.

Axis · axis-zyora · github.com/Zyora-Dev/axis

Reproduce: `modal run engine/modal_7b_bench.py`

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